Estimating FLS Sampling-Composition Weights by Entropy Calibration

1. Objective

The objective is to estimate the component of the regional Farm Labor Survey (FLS) wage generated by idiosyncratic variation in the realized geographic composition of the survey. The procedure begins with expected county contributions based on external employment data, uses auxiliary FLS moments to infer deviations from those expected contributions, and then applies the inferred weights to county OEWS agricultural wages.

The field-and-livestock FLS wage used to set the AEWR is not used as a calibration target in the preferred estimator. It is reserved for subsequent validation and first-stage estimation.

Let:

- i index counties;
- r index AEWR regions;
- t index years;
- q index FLS quarterly reference weeks;
- $m = 1, \dots, M$ index auxiliary moments.

2. Target Weight

The relevant object is a county's effective contribution to the FLS wage estimator, not its literal share of questionnaires. At the farm level, the effective contribution depends on selection, response, expansion weights, and reported worker hours. A stylized county contribution is

$$\omega_{irt} \propto \sum_{f \in i} I_{ft} R_{ft} d_{s(f)t} H_{ft},$$

where:

- I_{ft} indicates that farm f was selected;
- R_{ft} indicates that it supplied a usable response;
- $d_{s(f)t}$ is its expansion and nonresponse weight in sampling stratum $s(f)$;
- H_{ft} is its reported worker hours.

Normalize these effective county weights so that

$$\sum_{i \in r} \omega_{irt} = 1.$$

The simplified regional wage estimator is

$$\widehat{W}_{rt}^{FLS} = \sum_{i \in r} \omega_{irt} w_{irt},$$

where w_{irt} is the agricultural wage associated with county i.

3. Expected County Contributions

Let E_{it} denote hired agricultural employment and H_{it} average worker hours. The preferred prior contribution is

$$a_{it} = E_{it} H_{it}.$$

Normalize within each AEWR region-year:

$$q_{irt} = \frac{a_{it}}{\sum_{j \in r} a_{jt}}.$$

If county hours are unavailable, use

$$q_{irt} = \frac{E_{it}}{\sum_{j \in r} E_{jt}}.$$

The empirical hierarchy for E_{it} is:

1. county QCEW NAICS 11 hired employment, including a documented treatment of suppressed cells;
2. BEA hired farm employment, if separately available;
3. existing BEA farm employment;
4. OEWS agricultural employment as a sensitivity measure.

The prior describes expected effective geographic representation. It need not equal one at the county level; only its regional sum must equal one.

4. Sampling-Composition Deviations

Write the realized effective weight as

$$\omega_{irt} = q_{irt} + u_{irt},$$

where u_{irt} is the deviation from expected geographic representation. Since both weight vectors sum to one,

$$\sum_{i \in r} u_{irt} = 0.$$

The regional wage can then be decomposed as

$$\widehat{W}_{rt}^{FLS} = \sum_{i \in r} q_{irt} w_{irt} + \underbrace{\sum_{i \in r} u_{irt} w_{irt}}_{\eta_{rt}},$$

where

$$\eta_{rt} = \sum_{i \in r} u_{irt} w_{irt}$$

is the sampling-composition shock.

The complete vector $\{u_{irt}\}_i$ is not observed. The entropy-calibration procedure estimates a minimum-divergence approximation using auxiliary FLS moments.

5. County Features and FLS Targets

For each auxiliary variable m, construct:

$$z_{mit} = \text{county-level value of auxiliary variable } m,$$

and

$$T_{mrt} = \text{corresponding published FLS regional moment.}$$

Collect the county features in the vector

$$\mathbf{z}_{it} = \begin{bmatrix} z_{1it} & \cdots & z_{Mit} \end{bmatrix}',$$

and the regional targets in

$$\mathbf{T}_{rt} = \begin{bmatrix} T_{1rt} & \cdots & T_{Mrt} \end{bmatrix}'.$$

Every target and county feature must have the same interpretation and units. For example, the FLS share of workers employed 150 days or more must be paired with a county share defined using the same numerator and denominator as closely as possible.

6. Standardization

Within each region-year, calculate the prior-weighted mean of feature m:

$$\bar{z}^q_{mrt} = \sum_{i \in r} q_{irt} z_{mit}.$$

Calculate its prior-weighted standard deviation:

$$s^q_{mrt} = \sqrt{\sum_{i \in r} q_{irt} \left(z_{mit} - \bar{z}^q_{mrt}\right)^2}.$$

Define the standardized county feature

$$\tilde{z}_{mit} = \frac{z_{mit} - \bar{z}^q_{mrt}}{s^q_{mrt}},$$

and transform the FLS target using the same center and scale:

$$\widetilde{T}_{mrt} = \frac{T_{mrt} - \bar{z}^q_{mrt}}{s^q_{mrt}}.$$

Features with negligible within-region variation should be removed. Redundant quarterly shares or other linearly dependent moments must also be dropped.

7. Exact Entropy Calibration

For a given region-year, suppress the r, t subscripts for clarity. The exact calibration problem is

$$\min_{\{\omega_i\}} \sum_i \omega_i \log\left(\frac{\omega_i}{q_i}\right)$$

subject to

$$\omega_i \geq 0,$$

$$\sum_i \omega_i = 1,$$

and

$$\sum_i \omega_i \tilde{\mathbf{z}}_i = \widetilde{\mathbf{T}}.$$

The objective is the Kullback-Leibler divergence of the calibrated weights from the prior weights. The constraints require the calibrated county distribution to reproduce the selected auxiliary FLS moments.

7.1 Exponential-tilting solution

The solution has the form

$$\omega_i(\lambda) = \frac{q_i \exp\left(\tilde{\mathbf{z}}_i' \lambda\right)}{\sum_j q_j \exp\left(\tilde{\mathbf{z}}_j' \lambda\right)},$$

where λ is an M-dimensional vector of tilting parameters.

The dual problem is

$$\min_{\lambda} \left\{ \log \left[\sum_i q_i \exp\left(\tilde{\mathbf{z}}_i' \lambda\right) \right] - \lambda' \widetilde{\mathbf{T}} \right\}.$$

Its gradient is

$$\nabla_\lambda = \sum_i \omega_i(\lambda) \tilde{\mathbf{z}}_i - \widetilde{\mathbf{T}},$$

and its Hessian is

$$\nabla_\lambda^2 = \mathrm{Var}_{\omega(\lambda)}\left(\tilde{\mathbf{z}}_i\right).$$

The problem can be solved using Newton's method, BFGS, or another convex optimizer. The implementation should use a log-sum-exp calculation to prevent numerical overflow.

7.2 Feasibility

Exact calibration exists only if the target vector lies in the convex hull of the county feature vectors:

$$\widetilde{\mathbf{T}} \in \operatorname{conv}\left\{\tilde{\mathbf{z}}_i : i \in r\right\}.$$

Even when a solution exists, targets near the boundary of the convex hull can produce extreme weights.

8. Penalized Approximate Calibration

Exact balance is unlikely to be appropriate for the main estimator because county analogues are measured with error, some values are interpolated between Census years, and the FLS targets have sampling error. The preferred estimator therefore allows approximate balance.

Let $\mathbf{Z}'\omega$ denote the calibrated auxiliary moments. Solve

$$\min_{\omega} D_{KL}(\omega \| \mathbf{q}) + \frac{1}{2\kappa} \left(\mathbf{Z}'\omega - mathbfT\right)' \mathbf{V}^{-1} \left(\mathbf{Z}'\omega - mathbfT\right)$$

subject to

$$\omega_i \geq 0 \qquad \text{and} \qquad \sum_i \omega_i = 1.$$

Here:

- $D_{KL}(\omega \| \mathbf{q})$ penalizes departure from expected weights;
- $\mathbf{V}$ records the relative uncertainty of the auxiliary moments;
- $\kappa > 0$ controls the tradeoff between moment fit and weight stability.

Smaller κ places greater priority on matching the FLS targets. Larger κ shrinks the calibrated weights toward the prior.

A corresponding penalized dual problem is

$$\min_{\lambda} \left\{ \log \left[\sum_i q_i \exp\left(\mathbf{z}_i'\lambda\right) \right] - \lambda'\mathbf{T} + \frac{\kappa}{2}\lambda'\mathbf{V}\lambda \right\}.$$

Published FLS coefficients of variation can be used to construct the diagonal elements of $\mathbf{V}$. Because cross-moment covariances are generally unavailable, a diagonal matrix is a practical baseline. Alternative correlations should be considered in sensitivity analysis.

The tuning parameter κ must not be selected to maximize the AEWR first stage. It should instead be selected using auxiliary-moment fit, weight stability, effective sample size, or leave-one-moment-out prediction.

9. Weight Bounds and Stability

Define the adjustment factor

$$g_{irt} = \frac{\hat{\omega}_{irt}}{q_{irt}}.$$

Large values indicate that calibration greatly increases a county's contribution relative to its expected contribution. Useful sensitivity restrictions include

$$0.2 \leq g_{irt} \leq 5$$

and the wider alternative

$$0.1 \leq g_{irt} \leq 10.$$

These are sensitivity values, not known features of the FLS design.

Calculate the conventional effective sample size

$$N_{rt}^{eff} = \frac{1}{\sum_{i \in r} \hat{\omega}_{irt}^2},$$

as well as the prior-relative effective sample size

$$N_{rt}^{eff,q} = \frac{1}{\sum_{i \in r} q_{irt} g_{irt}^2}.$$

A sharp fall in either measure indicates that the auxiliary targets require an implausibly concentrated geographic representation.

10. Missing County Data

The calibration sample should not discard counties merely because their OEWS wage is missing. The OEWS wage is applied after estimating the weights and need not determine membership in the calibration sample.

Missing auxiliary variables should be treated using one of the following documented procedures:

1. imputation from state, OEWS area, farm type, and neighboring years;
2. missingness indicators included in the feature vector;
3. partial calibration using only moments observed for the common county set;
4. aggregation to a geography with adequate common support.

The amount of prior weight associated with missing data must be reported for every region-year.

11. Estimating the Sampling-Composition Wage Shock

Let x_{it}^{OEWS} be the county OEWS field-and-livestock wage proxy. After estimating the weights without using the FLS field-and-livestock wage, calculate the expected-weight wage:

$$\widehat{W}_{rt}^{expected} = \sum_{i \in r} q_{irt} x_{it}^{OEWS}.$$

Calculate the calibrated wage:

$$\widehat{W}_{rt}^{sample} = \sum_{i \in r} \hat{\omega}_{irt} x_{it}^{OEWS}.$$

The inferred sampling-composition shock is

$$\hat{\eta}_{rt} = \widehat{W}_{rt}^{sample} - \widehat{W}_{rt}^{expected},$$

or equivalently,

$$\hat{\eta}_{rt} = \sum_{i \in r} \left(\hat{\omega}_{irt} - q_{irt} \right) x_{it}^{OEWS}.$$

A positive value means that the inferred realized composition places greater weight on high-wage counties than the expected composition. A negative value means that it places greater weight on low-wage counties.

12. Timing

The weights, auxiliary FLS moments, and OEWS wages must refer to the same FLS wage year. If the AEWR in year $t + 1$ is set using the FLS wage for year t, then

$$\hat{\eta}_{rt}$$

is aligned with

$$AEWR_{r,t+1}.$$

For specifications using changes, construct the shock first and then difference it:

$$\Delta \hat{\eta}_{rt} = \hat{\eta}_{rt} - \hat{\eta}_{r,t-1}.$$

Do not mix outcome-year weights with prior-year wage measures.

13. Hierarchical State Calibration

If state-level auxiliary FLS targets become available, preserve each state's expected regional mass:

$$Q_{srt} = \sum_{i \in s} q_{irt}.$$

Calibrate counties conditionally within the state subject to

$$\sum_{i \in s} \omega_{irt} = Q_{srt}.$$

The conditional entropy solution is

$$\omega_{irt} = Q_{srt} \frac{q_{irt} \exp\left(\mathbf{z}_{it}' \lambda_{st}\right)}{\sum_{j \in s} q_{jrt} \exp\left(\mathbf{z}_{jt}' \lambda_{st}\right)}.$$

This preserves relative state size. Normalizing every state to one and then combining states would incorrectly give equal regional mass to small and large states.

If state auxiliary targets are unavailable, calibration should occur directly within the AEWR region. States should not each be calibrated to the same regional target.

14. Uncertainty

The calibrated weights are estimated objects. A simulation-based uncertainty procedure is:

1. Draw each FLS auxiliary target from an approximation to its sampling distribution using its published standard error.
2. Bootstrap or otherwise perturb estimated county auxiliary variables.
3. Re-estimate the entropy weights for each draw.
4. Recalculate $\hat{\eta}_{rt}$.
5. Propagate this variation into subsequent first-stage and outcome estimation.

If only marginal FLS standard errors are available, begin with independent draws and then evaluate sensitivity to positive correlations among related quarterly targets.

15. Diagnostics

For every region-year, report:

1. the number of counties in the prior and calibration samples;

2. the prior mass associated with missing auxiliary variables and OEWS wages;
3. prior and calibrated values of every auxiliary moment;
4. standardized balance errors;
5. the tilting coefficients $\widehat{\lambda}_{rt}$;
6. the distribution of $g_{irt} = \widehat{\omega}_{irt}/q_{irt}$;
7. the maximum and minimum adjustment factors;
8. conventional and prior-relative effective sample sizes;
9. whether the exact target vector lies in the counties' convex hull;
10. the expected and calibrated OEWS wages;
11. the inferred shock $\widehat{\eta}_{rt}$;
12. sensitivity to moment selection, tuning parameters, and weight bounds.

16. Recommended Baseline Estimator

The initial estimator should use:

- employment-based prior weights;
- four or fewer auxiliary constraints with clear county analogues;
- standardized county features;
- penalized rather than exact entropy calibration;
- no FLS wage variables as calibration targets;
- an effective-sample-size floor;
- multiple reasonable weight-bound specifications;
- tuning selected using auxiliary fit and stability rather than first-stage strength.

The contemporaneous exact calibration to the FLS field-and-livestock wage can be retained as a descriptive upper-bound exercise. It should not be treated as the preferred excluded instrument because the AEWR-setting wage enters the resulting weights by construction.